Computers Cannot Imagine: The Fundamentals of Synthetic Image Generation (Workshop)

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Generative AI, and more specifically tools such as DALL-E and Midjourney, have introduced synthetic image generation capabilities that appear both magical and transformative. Anyone can, by simply describing a concept in plain language, conjure into existence an image—seemingly of anything one might imagine.

Rationally, we understand that this is not in fact *magic*, but instead the output of clever programming and statistical tools operating over an inconceivably large amount of digitized visual culture. But without a clear understanding of the mechanisms behind these tools —and an intuitive sense of both their true capabilities and their limitations— we cannot effectively take advantage of their capabilities, nor properly position them within the larger conversation around the human impulse to create images and use them to communicate with each other that takes place in across any number of humanities disciplines.

In this workshop, we will provide a framework for conceptualizing how contemporary synthetic image generators work, from both theoretical and technological perspectives. We will also demystify the process of image generation by providing analogies and accurate mental models for what is happening within the system. We will equip participants both to take up these tools productively in their own research and also to explain what is happening "under the hood" in plain, but not overly simplistic, language. It is our hope that, with a greater awareness of the technical realities behind the synthetic image generators, the community of humanists can begin to open up a broader conversation about how these technologies can help us better understand what we can and cannot say about the human experience of creativity and image consumption.

This workshop will consist of a combination of hands-on activities focused on the exploration of image generation tools, expert presentations about the underlying technologies and processes that underpin synthetic image generation, as well as group interactions and discussion. It will also include practical advice for how to incorporate discussion of these tools into academic research and pedagogy.

The workshop is targeted at scholars who work with visual culture in their research and teaching who also seek a deeper understanding of how generative AI functions as a technology. This workshop will be appropriate for individuals at all levels of technological experience, but particularly useful for those with a clear understanding of the traditions of studying visual culture as an act of human creativity, but without sophisticated programming skills. If you are familiar with the idea of Markov chains but have never explored what they are—this workshop is for you. If you are familiar but are curious how they relate to image generation, this workshop is also for you. And If you have never heard of them at all, we'll introduce you to them you!

A laptop will be needed, but no additional software or hardware will be required.

The Workshop Leaders

Alison Langmead ( ) holds a joint faculty appointment at the University of Pittsburgh between the Dietrich School of Arts and Sciences and the School of Computing and Information. She serves as the Director of the Visual Media Workshop (VMW), a digital humanities lab located in the Department of the History of Art and Architecture that focuses on the investigation of material and visual culture in an environment that encourages technological experimentation. Her publications have recently appeared in *Digital Humanities Quarterly*, *The Routledge Companion to Digital Humanities and Art History*, *Technology and Culture*, and the *International Journal for Digital Art History*. She has been the Principal Investigator for projects funded by the National Endowment for the Humanities, the J. Paul Getty Foundation, and the Andrew W. Mellon Foundation, among others.

David Newbury ( ) is the Senior Director, Public Technologies at the J. Paul Getty Trust, where he provides strategic direction and leads teams responsible for imaging, collections information, digital platforms, and computational heritage.

David combines deep expertise in software de-

velopment and image analysis with a deep understanding of the practical impact of technology in our understanding of visual culture as understood from the perspective of memory institutions. Much of his work revolves around the practical application of technology to visual culture, with an emphasis on human-computer interaction, multi-disciplinary collaboration, and the application of semantic representations of cultural information for public access.

Half-Day Workshop Schedule Outline (180 minutes total)

- Introduction to the Workshop and Warm-up Exercise (30 minutes)
  - Introduction to the Workshop and Workshop Presenters (5 minutes)
  - "Open Play Time" with a synthetic image generator (ChatGPT/DALL-E)) to allow participants to gain a personal feel for how these technologies behave (15 minutes)
  - Share-Out Time for the participants' experiences and results (10 minutes)
- Presentation: Introduction to the Synthetic Image Generating Technologies (35 minutes)
  - Langmead and Newbury will present an overview of the technological underpinnings behind the synthetic image generators, focused on situating these systems in a context that scholars of visual culture can most easily understand.

Break (10 minutes)

- Hands-on Workshop (60 minutes)
  - What are Markov chains, and what do they have to do with synthetic image generation? An introduction to the basic statistical underpinnings of these technologies. (5 minutes for introduction; 15 minutes for the exercise)
  - What, precisely, can users control when they work with the synthetic image generators? An introduction to prompt engineering, but also selected features of the Flux tool by Black Forest Labs (5 minutes for introduction; 15 minutes for the exercise)
  - "Computer, make me an image of something NOT on the Internet!" An introduction to the ways to ask these systems to show the limits of their "vision." (5 minutes for introduction; 15 minutes for the exercise)

Break (10 minutes)

Structured, Group Share-Out Conversations (35 minutes)

- Central questions the group will be asked to consider together will include:
  - What did you learn today? What did you now see you did not know coming into this workshop?
  - What has changed your mind and how did it happen?
  - Would you like to take this into your classroom? If so, how are you thinking of doing so?